two degrees of freedom due to vibrational motion[one due to kinetic energy of vibration and the other is due to potential energy] (Figure 9.5c). So totally there are seven degrees of freedom.

$$f = 7$$

**Examples:** Hydrogen, Nitrogen, Oxygen.

### 9.3.4 Triatomic molecules

There are two cases.

**Linear triatomic molecule**

In this type, two atoms lie on either side of the central atom as shown in the Figure 9.6

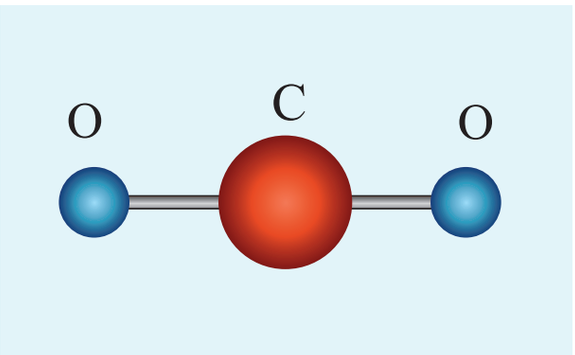



**Figure 9.6** A linear triatomic molecule.

Linear triatomic molecule has three translational degrees of freedom. It has two rotational degrees of freedom because it is similar to diatomic molecule except there is an additional atom at the center. At normal temperature, linear triatomic molecule will have five degrees of freedom. At high temperature it has two additional vibrational degrees of freedom. So a linear triatomic molecule has seven degrees of freedom.

**Example:** Carbon dioxide.

**Non-linear triatomic molecule**

In this case, the three atoms lie at the vertices of a triangle as shown in the Figure 9.7

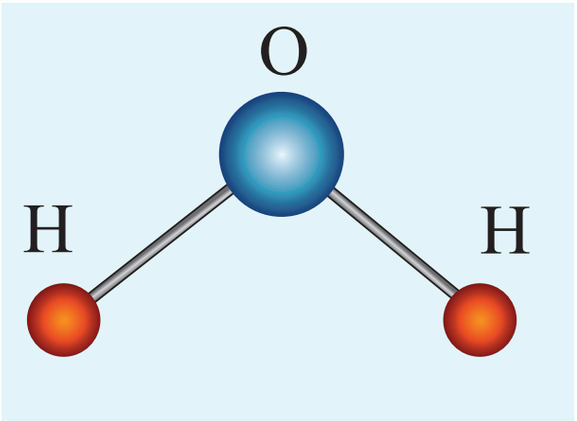



**Figure 9.7** A non-linear triatomic molecule

It has three translational degrees of freedom and three rotational degrees of freedom about three mutually orthogonal axes. The total degrees of freedom, f = 6

**Example:** Water, Sulphurdioxide.

## 9.4 LAW OF EQUIPARTITION OF ENERGY

We have seen in Section 9.2.1 that the average kinetic energy of a molecule moving in x direction is $\frac{1}{2}m\overline{v_x^2} = \frac{1}{2}kT$ .

Similarly, when the motion is in y direction, $\frac{1}{2}m\overline{v_y^2} = \frac{1}{2}kT$ and for the motion along z direction, $\frac{1}{2}m\overline{v_z^2} = \frac{1}{2}kT$ .

*According to kinetic theory, the average kinetic energy of system of molecules in thermal equilibrium at temperature T is uniformly distributed to all degrees of freedom (x or y or*



UNIT-9(XI-Physics_Vol-2).indd 175 26-03-2022 17:47:20